# CHAPTER- II

# PROGRAMMING FUNDAMENTALS

## STRUCTURE OF JAVA:

- Java is a case sensitive language. Upper and lower case plays a different role.

Example code:

```
public class Example{
    public static void main( String [] args){
            System.out.print("welcome");
    }
}
```

Here:

- ✓ **Example** -is a class name the file should be same as a class name.
- ✓ **public static void main(String [] args)** – main method.
- ✓ **System.out.print("welcome"); -** is an output visible in the screen.

## JAVA SYNTAX RULES:

- The file name should be same as a class name.
- The curly braces {} defines the block of code.
- Every statement in the program should be ends with semicolon(;).
- Java is a case sensitive so the variable names should be written carefully. Eg: num != Num.
- Variable and method name should be starts with lowercase.
- Class name should be starts with Uppercase.

## COMMENTS IN JAVA:

- Comments are used for the understandable to the user. It doesn’t display or defines error in the code.
- **Types:**
  - Single line (//)
  - Multi line(/* … */)

## DATA TYPES:

- Data types were used to define the type of the value used in the program.
- **Types:**
  - **int**- integer value (eg: int a=10;)
  - **char**- single character (eg: char ch= ‘A’;)
  - **double**- decimal value (eg: double a = 10.05;)
  - **string**- text (eg: string name= “abi”;)
  - **boolean**- true/false (eg: Boolean isEligible= true;)

## PRIMITIVE DATA TYPE:

- Primitive data type has a fixed size and specific purpose.
- There are 8 primitive data types are available.
- Data types: byte, short, int, long, float, double, char and Boolean.

## NON- PRIMITIVE DATA TYPES:

- Non- primitive data types are created by the programmers.
- It doesn’t have the fixed size.
- Data types: string, array, object, class and interface.

## VARIABLES IN JAVA:

- Variables is a container that stores the value and allocate memory space for the value.

**Syntax:**

```
datatype variablename= value;
```

**Example:**

```
public class Example{
        public static void main(String [] args){
                int a=10;
        }
}
```

## RULES FOR DECLARING A VARIABLE:

- Variable should be start with a character, $ or _.
- Variable name shouldn't be the java keywords.
- Variable name is case sensitive.
- Numbers are not used in the beginning of the variable name.

## TYPE CASTING:

- Type casting means converting one data type into another data type.
- **Types:**
    - Widening (Automatic) Type Casting- small to large
    - Narrowing (Manual) Type Casting- large to small

## OPERATORS IN JAVA:

- Operators are the special symbols that used to perform operations on variables and values.
- **Types:**
  - Assignment
  - Arithmetic
  - Logical
  - Relational
  - Unary
  - Ternary
  - Bitwise

## OUTPUT IN JAVA:

- To print and display the value in the screen.
- Syntax:

```
System.out.print("hello");
System.out.println("hi");
```

## INPUT IN JAVA:

- To get the input from the user.
- It uses a Scanner class.
- Example:

```
Scanner sc= new Scanner(System.in);
```

## JAVA SCANNER INPUTS:

- sc.nextLine() – string
- sc.nextInt() – int
- sc.nextFloat() – float
- sc.nextByte() – byte
- sc.nextBoolean() – boolean
- sc.nextDouble()- double
- sc.nextShort() – short
- sc.nextLong() - long